

READ THIS

- Questions are from these four sections \\\

What's your design brief/opportunity

- Explain it well and solid

Questions about diverging

Question about prototype

- Which prototype do you think is the worst and why?
- Now what's the best and why
- Question could be anything related to the prototypes

Did you rescope

- We have to have rescoped, make something up
- If we don't rescope, its saying the design brief was perfect, which they are definitely not based on the marks they gave it (they marked them already)
- Why did you rescope, what did you rescope, based on what did you rescope

What research have you done

- Any type of research
- Reference design

1. Design brief interpretation: you have to decide how to approach your chosen design brief as a design team. Whether you wrote the brief yourselves or are choosing another one, you may need to re-scope or reframe, and you will certainly need to refine the original opportunity. This likely will involve **research** to enable you to:
 - revise the stakeholders, defining them more precisely,
 - narrow the scope to a subset of the earlier scope,
 - **focus and refine the objectives into a set of requirements and evaluation criteria.** This may require adding/dropping a goal/objective, making objectives more precise, etc.
 - redefine metrics so that you can actually test your possible design alternatives (once you have them).
 2. Process: in studio during the week of 2025-10-13, you will be guided through an initial cycle of idea generation and divergence. This will likely be done rapidly and without research. By Alpha, you should be able to show **multiple iterations of diverging with research** in order to demonstrate that you have approached the opportunity from multiple angles.

You should also be prepared to demonstrate your team's strategy for working and decision making. There is no "right" answer here, but you should be able to show and explain how your team develops ideas, makes decisions, and ensures equitable contributions from all members.
 3. Design concepts: at this stage, we expect you to have at least **$n+1$ concepts** that you are considering, where n is the number of members on your team. Good concepts are those that are legitimately worth your consideration, and that appear, at least for now, to meet your requirements. At this stage, the representations of many of these concepts can be very sketchy (literally, they can be sketches).
 4. Next steps: at this stage, you should be able to tell the teaching team what your **biggest questions** about each of your design concepts are (but NOT the answers to these questions!). You might be able to identify which metric(s) will be critical (i.e., which ones will really tell whether your idea will work). You might even be able to say how you could "proxy" a test for these metrics given your available time and equipment.
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Some TA's don't direct questions to a specific person or look at a specific person, they address the team

- IT LOOKS BAD IF WE LOOK AT EACH OTHER
- OR TALK OVER EACH OTHER

TIPS

- Say WE never I
 - "WE did this, WE realized this, WE thought this"
- Speak formally, no like, um, etc
- They like cohesion
 - EVERYONE IS ON THE SAME PAGE
 - Wear the same colours WE ARE WEARING BLACK TOP AND BOTTOM
 - DONT TALK OVER EACH OTHER
 - Everyone has to talk EQUALLY
- They care about group dynamics ASK ABOUT HOW THE TEAM WORKED TOGETHER
 - Glaze each other up when talking about the work experience

Questions

- Most important requirement
- Least important requirement
- Most used diverging tool
- Least used diverging tool
- Favorite diverging tool
- Least favourite diverging tool
 - The tool didn't help with brainstorming but helped bond as a group, etc
 - SOMETHING POSITIVE
- Summarize your design brief in two sentences
- Summarize your opportunity in two sentences
- Demonstrate how to use each prototype
 - MAKE IT ENGAGING
- Which prototype do you think will work the best

To Do

- Finish editing design brief
- Finish making the requirements/evaluation criteria

Figure out which prototype is the best and which is the worst, why and based on what criteria?

- Best: the dummy marker
 - Because it does not restrict the range of angles of the user's grip and allows them to keep writing in their comfortable positions (1.2) and accommodating for various grip styles on the pen(1.3). It is also able to rest comfortably in a 4.9cm by 5.5cm marker tray for convenience (2.1). Additionally, there are no portions of the design that obstruct the writing area within the user's line of sight (from eye level to hand position)(2.2). It does not produce any sound disturbance to room(2.3). There is no unintentionally smudged ink at the written location due to the position of the real marker placed behind the hand(3.1, 4.1). There will be no nicks made on the whiteboard as marker strokes are written with a white board marker(5.1). No stains are left on the whiteboard surface as whiteboard markers are dry erase(5.2).
- Second place: roller?
- Worst: platform
 - Might induce wrist strain as the hard rigid platform might restrict the range of angles of wrist movement/grip styles(1.2, 1.3). The design will need disassembling the platform to fit on the marker tray(2.1). The platform might cause unintentionally smudged ink(3.1).

What diverging tools did we use? Evidence of the diverging process.

- We are supposed to have multiple iterations of diverging with proof
 - Used the SCAMPER model, freewriting, wishing.
 - Evidence is on chart paper and printed paper
- Combinations of free writing and wishing is the best
- Best to worst: wishing & freewriting, 635, SCAMPER

Which diverging tool was the best? Worst? Why?

- The best was the wishing/free writing
 - Similar to free writing in class, which we had practice with
 - Most of our ideas come from here
- Worst was SCAMPER

What did we rescope and why did we rescope

- Rescoped requirement of no accidental contact with surface of whiteboard into no accidental contact with marker ink. because

What research did we do

Reference Designs

- Airhockey table (fan)

- Drawing compass (dummy marker)
-

Chemical formula of permanent markers (various alcohols such as isopropanol)→ chemicals (isopropyl/ethanol alcohol, acetone, acetic acid) that could be solvent(dissolving the ink and breaking its cohesive forces with whiteboard).

- **Researched used when making our prototypes**

Which requirement is the most important? Which is the least important?

- 3.1 → no smudge writing

How did we ensure equitable contributions from each member?

- diverging /converging together
- Dividing work and

Biggest questions about each design concept:

- Dummy marker
 - What's the optimal angle?
 - What material can we use on the bottom of the marker that the user holds to not damage the whiteboard?
- Platform
 - What size of platform will accommodate all grip styles?
 - How strong does the platform have to be?
 - How would we make the platform detachable?
- Roller
 - What is the coefficient of static friction between materials and the whiteboard?

Actual Opportunity Brainstorming

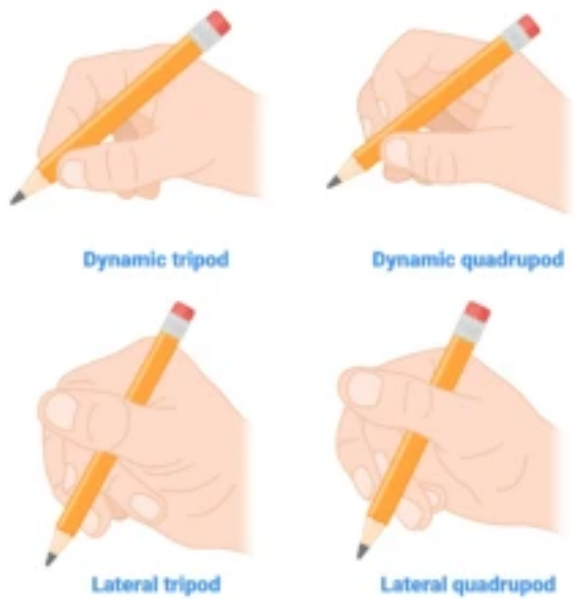
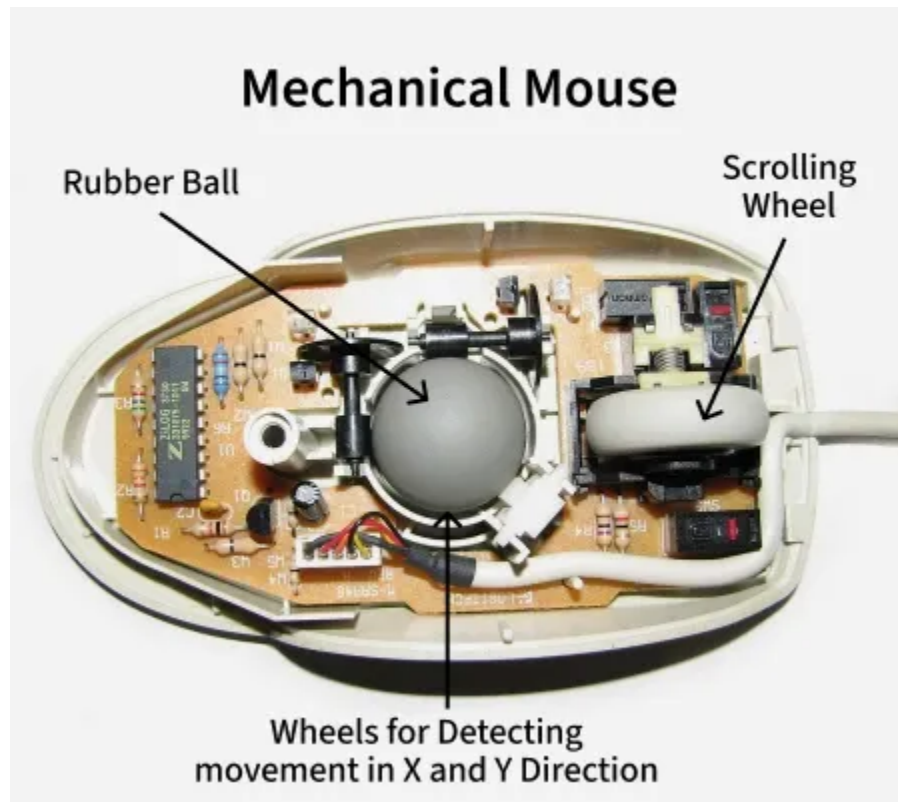
Kind of plausible ideas (Inspired by Wishing Diverging Method and free writing)

1. Improved material (stain + sweat resistance) for the glove so that there is no smear at all (whatever the criteria is for an improvement on the smearing)
2. Quick drying markers that requires special eraser to be erased so hands can't erase the marker -> change the chemical composition of the ink
 - a. Reference design: Eraseable pens
3. Repelling magnet → magnetic levitation
 - a. Could also use electromagnets
4. Air cushion attachment
 - a. Get exact values later from reputable sources but roughly the following:
 - i. The average person exert 1.5N of force on the table while writing at a normal speed
 - ii. A rough estimate of the area of contact is 30cm^2
 - iii. A rough pressure exerted by the hand is 50 Pa
 - iv. Figure out how much air flow rate is needed to exert an equivalent opposite pressure of 50 Pa
 - b. The fan cannot be too loud nor can the attachment be too heavy
 - c. Porous surface at the bottom for which the air can go through
5. Plastic attachment to hand to contact board
 - a. Rolling ball?
6. Digital board (can just busy)
7. Make a supporting section/anchor/platform on the marker that the palm rests on instead of the board.
8. Bent marker that writes to the left of it instead of where it's pointed?
 - a. Can be held just like a normal marker
 - b. Solution one: over the hand attachment
 - c. Solution two: around the hand attachment
9. Dummy marker (basically have a second marker to the left of the real one that is the one that actually draws)

Absurd ideas (Wishing Diverging Method):

1. I wish people would just write with the right hand
2. I wish I could change english to be written from right to left and force all righties to write with left hand
3. I wish I could invert the physical properties of the universe so that left now becomes right for specific people
4. I wish I could write a sentence from the end to the start
5. Reverse time locally on the board/delay time on the whiteboard
6. Biomedical engineering so that all babies are born right handed/ambidextrous
7. Marker that communicates with a second robot marker. The real marker won't draw anything, but it'll tell the robot marker to draw that thing a few seconds after the real marker draws it -> maybe a dummy marker physically attached instead of wirelessly connected?
8. Hire another person to draw for you
9. Make schools to force left handed people to write right handed
10. Transfer what wants to be drawn in brainwaves to the whiteboard
11. Talk to an AI model that controls a robot marker

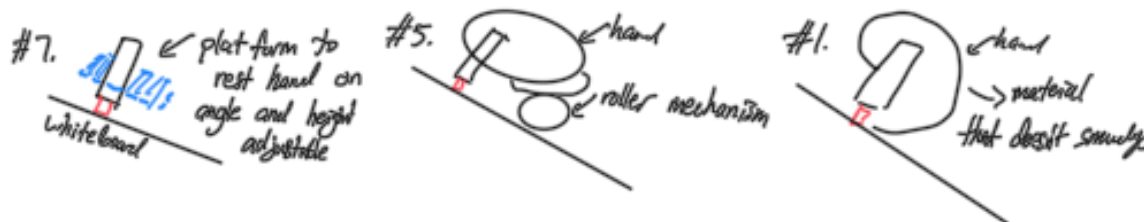
Reference for rolling wheel design



Preconverge (n+1):

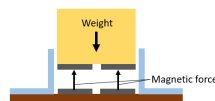
1. Quick drying markers that requires special eraser to be erased so hands can't erase the marker -> change the chemical composition of the ink
 - a. Reference design: Eraseable pens (heat erase) + heating eraser (60 c)
 - b. Permanent ink + alcohol spray eraser
2. Dummy marker (basically have a second marker to the left of the real one that is the one that actually draws)
 - a. Reference design: compass drawing tool
3. Fan thing
 - a. Reference design: air hockey machine
4. Make a supporting section/anchor/platform on the marker that the palm rests on instead of the board.
 - a. Reference design: Gimbal, camera stabilization
5. Rolling ball hand support -> sphere with rollers on the sides in all three axis like the mechanical mouse.

Drawings:

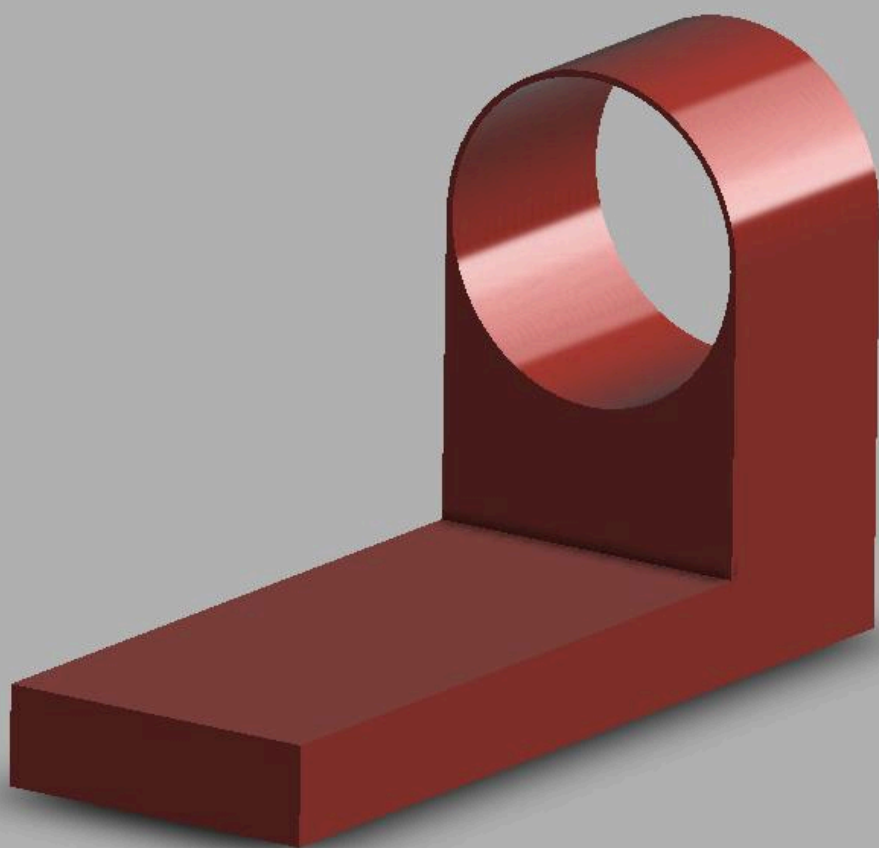


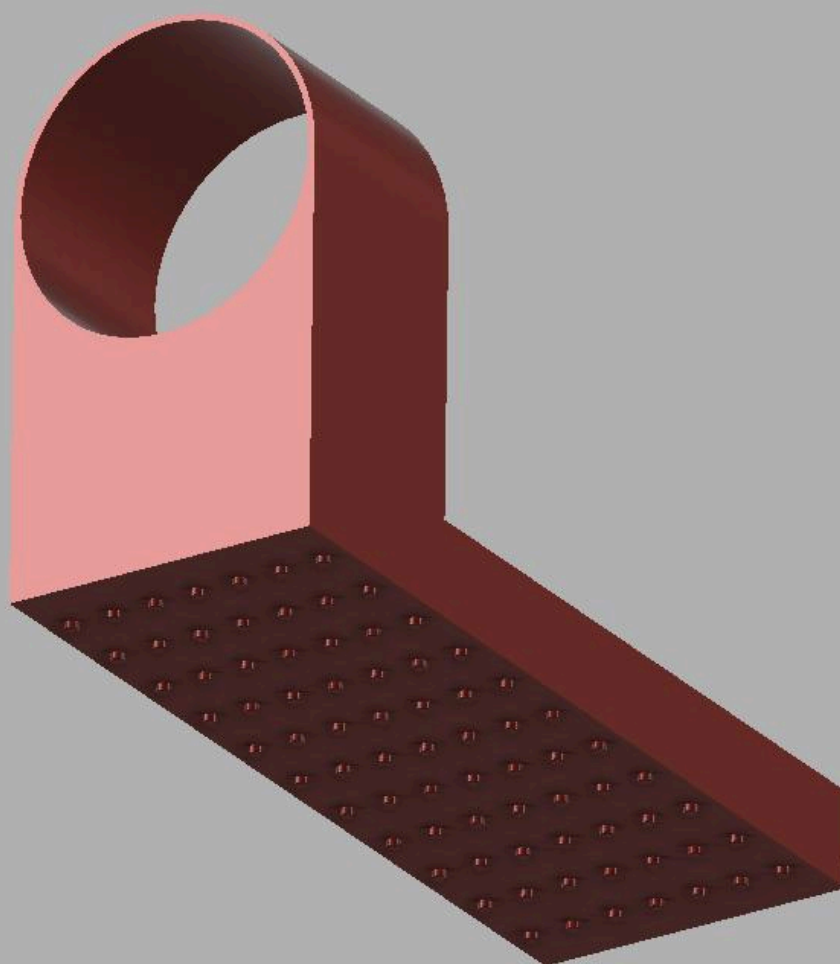
Reference Designs

- Air Hockey Table levitation mechanism
 - <https://www.gettrampoline.com/air-hockey/how-air-hockey-table-work-parts-physics/>
- Air Hockey DIY design:
 - <https://www.instructables.com/Desktop-Air-Hockey-Table/>
 - Uses computer fans
- Magnetic levitation

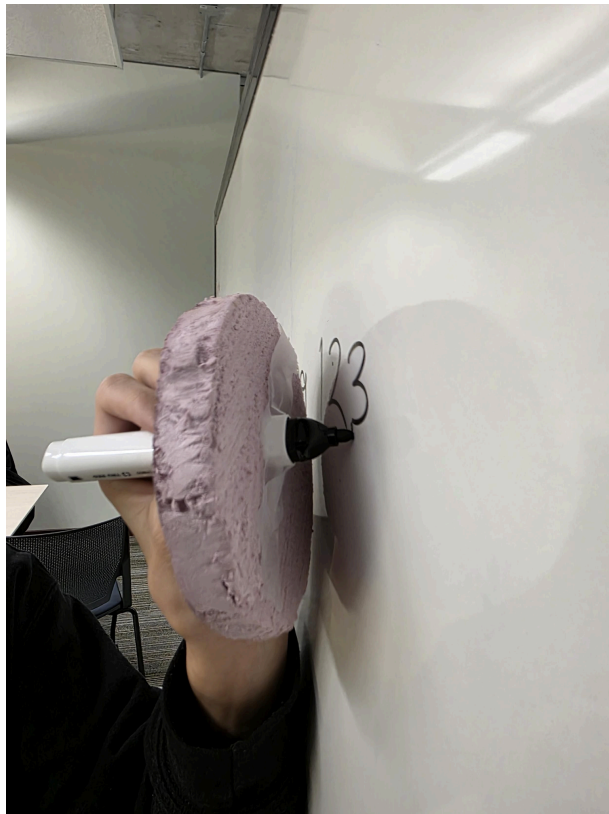
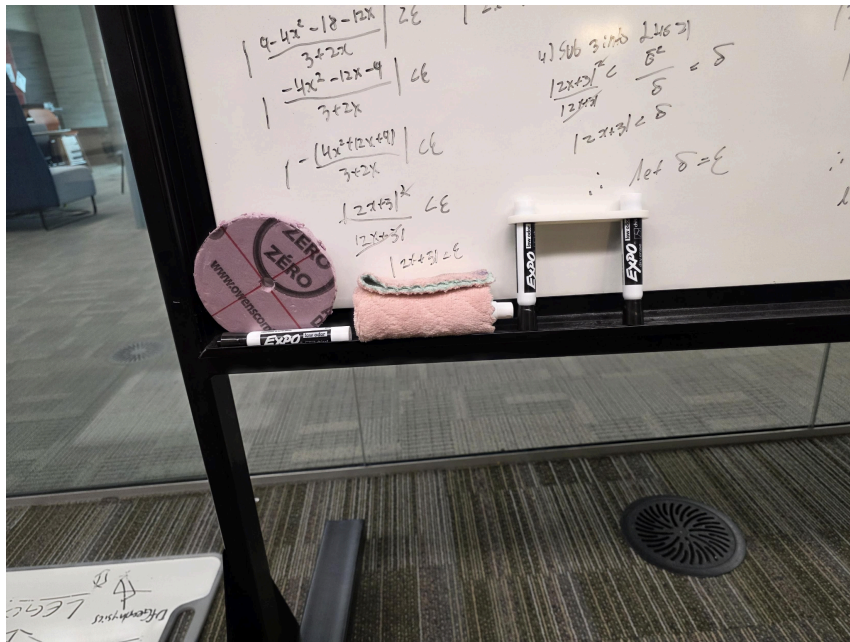


- <https://coertvonk.com/physics/electromagnetism/magnetism/levitation-30367>

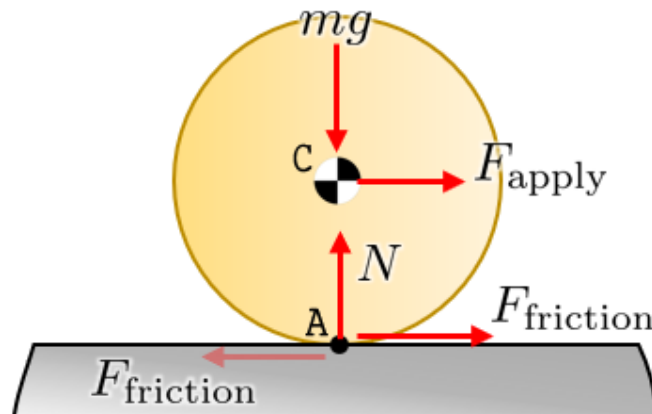




https://www.libertygames.co.uk/store/air_hockey_tables/buying-advice/what-is-air-hockey/?srsltid=AfmBOoq6nGj-UC6qztCuojMsVg6kT71pL1uUNangqxERKhylUpTudioV -> says that pucks float on air hockey table, not touching the table.



Mathematical Model for Roller Design



Let F_s be the force of static friction

$$F_s = \mu_s F_n$$

If $F_{\text{apply}} < F_s$, no slipping will occur

Materials like polystyrene have $\mu_s = 0.5$ [1][2], and F_N is estimated to be 10 N. $F_s = 0.5 * 10 \text{ N} = 5 \text{ N}$.

Average handwriting speed maximum of around 15 words per minute [3][6] = 1 word/4 seconds. Average word has 4.7 characters in it [4][5], and if they are writing in font size of 10 cm in height, then we have approximation of $4.7 * 10 * 2 = 94 \text{ cm} / 4 \text{ seconds}$, or $23.5 \text{ cm} / \text{second} = 0.235 \text{ m/s}$. Assuming a time of acceleration of 0.1 seconds, we have $0.235 \text{ m/s} = a\Delta t$, so $a = 0.235 / 0.1 = 2.35 \text{ m/s}^2$ of acceleration for F_{apply}

$$F_{\text{apply}} = ma = 2 \text{ kg} * 2.35 \text{ m/s}^2 = 4.7 \text{ N}.$$

Since $4.7 \text{ N} < F_{s\text{max}}$ of 5 N, this design has the potential to work

We would use ASTM D1894 to measure the coefficients of static friction of materials to be tested. <https://www.instron.com/en/testing-solutions/astm-standards/astm-d1894/>

ISO 8295-1995 could also be used

<https://cdn.standards.iteh.ai/samples/20760/7ef70177b6af4ae4bc307a612858d017/ISO-8295-1995.pdf>.

[1] https://www.engineeringtoolbox.com/friction-ths-d_778.html

[2] https://www.engineersedge.com/coefficients_of_friction.htm

[3]

https://tctecinnovation.com/blogs/daily-blog/comparing-handwriting-speeds-among-different-age-groups?srltid=AfmBOopowQraPUd5GZkoSOpLu1qFuyz9BbhOJRIjf3HYk7_gS24Engb9

[4]

https://www.researchgate.net/figure/Average-word-length-in-the-English-language-Different-colours-indicate-the-results-for_fig1_230764201

[5] <https://norvig.com/mayzner.html>

[6] https://www.montgomeryschoolsmd.org/siteassets/district/departments/hiat/resources/handwriting_speeds.pdf

Goals:

Design must not affect the user's baseline writing ability

Design must be comfortable to use and have around the commons.

Ensure no unintentional movement of the ink on the surface of the whiteboard occurs

Ensure no ink residue is left on the hand after use of the device

Design must not damage the whiteboard

Requirements:

1. ~~Cannot inhibit the users baseline DASH-2 score by more than 15 points~~
2. ~~Ensure there is no risk of strain injury caused by wrist extended by $>90^\circ$, wrist radial abduction, $>20^\circ$, and wrist ulnar abduction, $>30^\circ$, outlined in ISO 11226-2000~~
3. ~~Accommodate for various grip styles on the pen, namely tripod and quadropod grip styles~~
4. ~~Design width of at most 4.9cm.~~
5. ~~Design colours are distinguishable, colour difference between the design and the marker rail (silver, #CoCoCo) must have a perceptual $\Delta E^* \geq 20$ according to ISO/CIE 11664-~~
6. ~~Reduce disturbance by maintaining the common room's sound pressure level to under 60 dB.~~
7. ~~No contact between design and ink that removes ink unintentionally~~
 - a. ~~USED TO BE: No accidental contact between design and surface of the whiteboard **within** 27.73s [19] of the ink being applied.~~
 - b. ~~ADD REASONING~~
8. ~~No accidental contact between design and surface of the whiteboard **after** 27.73s of the ink being applied.~~
9. ~~Zero points of contact between the hand of the user and the surface of the whiteboard.~~
10. ~~Any nicks on the whiteboard surface must be "no greater than 0.005" (0.01270 cm) in depth and 0.010" (0.0254 cm) in width."~~
11. ~~Any staining from the design should have $\Delta E^* \leq 1.9$ with whiteboard surface.~~

Potential Opportunities

1. An Acoustically Optimized Depressurization Device.pdf
2. Backpack Weight Reduction System for Bicycles.pdf
3. Bathroom Stall Storage.pdf
4. Commuter Students Studying on the Go.pdf
5. Design Optimization of a Hand-Held Broom for Users with Mobility Restrictions.pdf
6. Difficulty in Peeling Bananas.pdf
7. Inconvenience of Bike Security.pdf
8. Inconveniences of Near-Empty Pump Dispensers.pdf
9. Individually Adhering and Aligning Posters Mechanism.pdf
10. Left-Handed Whiteboard Use in EngSci Common Room.pdf
 - a. Improved material (stain + sweat resistance) for the glove so that there is no smear at all (whatever the criteria is for an improvement on the smearing)
 - b. Quick drying markers that requires special eraser to be erased so hands can't erase the marker
 - i. Reference design: Eraseable pens
 - c. Repelling magnet → magnetic levitation
 - d. Air stream air hockey thing
 - e. Plastic attachment to hand to contact board
 - i. Rolling ball?
 - f. Digital board (can js buy)
11. Making Fitness Accessible - Improving the Ergonomics and Accessibility of Weight Loading Systems in Gyms.pdf
12. Reducing Light and Noise for Sleep in a Shared Dorm Room.pdf
13. Tool for Removing Hair from Casters on Rolling Chairs.pdf

Eliminating Opportunities

Addressing the Inconvenience of Carrying a Spring Jacket in Dynamic Urban Environments.pdf

Addressing the Presence of Excess Backpack Strap Length.pdf

An Acoustically Optimized Depressurization Device.pdf

An Opportunity to Enhance the Kawasaki Ninja 400 Motorcycle Ignition Slot Interface.pdf

Backpack Handling on TTC Subway Trains During Peak Hours of Service.pdf

Backpack Weight Reduction System for Bicycles.pdf

Backpacks Slip Off the Backs of Robarts Library Chairs.pdf

Bathroom Stall Storage.pdf

Better Water Bottle Washing and Filling at Chestnut Residence.pdf

Broken or missing locks in UTSG washrooms.pdf

Challenges in One-Handed Keycard Retrieval at Chestnut Residence.pdf

Chestnut Upper Dryers Accessibility Issue.pdf

Cleaning Dirty Keyboards.pdf

Commuter Students Studying on the Go.pdf

Creating Better Pre-sleep and Awakening Windows for Engineering Students.pdf

Design of a Reliable Wake-Up Device to Prevent Oversleeping in Engineering Science Students .pdf

Design Optimization of a Hand-Held Broom for Users with Mobility Restrictions.pdf

Difficulty Biking in the Rain.pdf

Difficulty in Peeling Bananas.pdf

Doffing Disposable Gloves.pdf

Enhanced Eating Experience of Wrapped Takeout Food.pdf

Enhancing Plug Stability for Portable Charger Connections.pdf

Ergonomic Grocery Bags.pdf

From Lecture to Lecture - Footwear for Rainy Campus Days.pdf

Getting a Bike To and/or From Chestnut's Bike Parking Lot .pdf

Hard-To-Reach Backpacks.pdf

Improving Ergonomics and Comfort of Napping in EngSci Common Room Chairs.pdf

Improving Rolling Chair Safety.pdf

Inconvenience of Bike Security.pdf

Inconveniences of Near-Empty Pump Dispensers.pdf

Individually Adhering and Aligning Posters Mechanism.pdf

Investigating Inadequate Paintbrush Cleaning for the Tired EngSci Artist.pdf

Lack of Security for Personal Belongings in the Engineering Science Common Room.pdf

Laptop Hinge.pdf

Laundry Process Inefficiency at Chestnut Residence.pdf

Leaking Cafeteria Lunchboxes .pdf

Left-Handed Whiteboard Use in EngSci Common Room.pdf

Left-Side Toilet Paper Dispenser Placement in New College Accessibility Washrooms.pdf

Lighting in Chestnut Residence.pdf

Making Fitness Accessible - Improving the Ergonomics and Accessibility of Weight Loading Systems in Gyms.pdf

Maximizing Ink Usage of Whiteboard Markers.pdf

Music Stand for Convenient and Uninterrupted Practice .pdf

On the Issues Associated with Locks in Jack-and-Jill Bathrooms at the Oak House Residence.pdf

Optimising Backpack Cable Storage .pdf

Optimizing Key Holder Efficiency for First-Year U of T Students.pdf

Preserving Bicycle Brake Reliability Under Wet Conditions.pdf

Preventing Aqueous Residues' Deposition onto Tabletop Surfaces.pdf.pdf

Putting an Issue to Rest - Sleeping on the TTC.pdf

Reducing Light and Noise for Sleep in a Shared Dorm Room.pdf

Reducing Noise of Ice Cubes In Reusable Water Bottles.pdf

Removal of Liquid for Disposable Cup Recycling.pdf

Soggy Sandwiches.pdf

Splashes from Chestnut Water Refill Stations.pdf

Suitcase to Minimize Jostling While in Transit.pdf

Tall Library Bookshelves.pdf

The Hunger Games - Improving SF Vending Machine Product Retrieval Accessibility.pdf

The Need for Fast-Drying Umbrellas.pdf

The problem of backpacks causing pain.pdf

Theft in the Chestnut gym.pdf

Tool for Removing Hair from Casters on Rolling Chairs.pdf

Unnecessary Hassle of Taking Supplements in the Morning.pdf

Wear of Shoe Heels Caused by Heel Dragging.pdf

Wet Folding Umbrellas Dripping Water onto the Floor.pdf

Wet umbrella causing indoor mess.pdf

Tab 4

What is a diverging tool that helped you overcome a cognitive anchor?

Why and how did it disrupt your thinking?

Tab 7

